Structural Deficits in the Silver Market:

Drivers, Risks, and Industrial Implications *

Dhruv Chhabra

April 1, 2026

1 Current Landscape

The silver market has entered a phase of unprecedented volatility, spot prices hitting an all-time high of \$121.60/oz in January 2026 before a macro-driven correction. Despite recent margin hikes by the CME Group, aimed at cooling the market, institutional and sovereign demand continues to absorb supply at record rates. This rally is no longer driven by speculative traders but by a fundamental supply deficit projected to entering its 6th consecutive year in 2026. This dynamic, partially amplified by China's Export Licensing restrictions in January, shifted the market from demand-led to a supply-constrained growth. For solar PV producers, the binding risk is no longer just price, but the loss of physical access to the metal, threatening both near-term EBITDA and 2030 production targets.

2 Underlying Drivers

Silver's valuation framework is shifting and it's driven by these four concurrent forces.

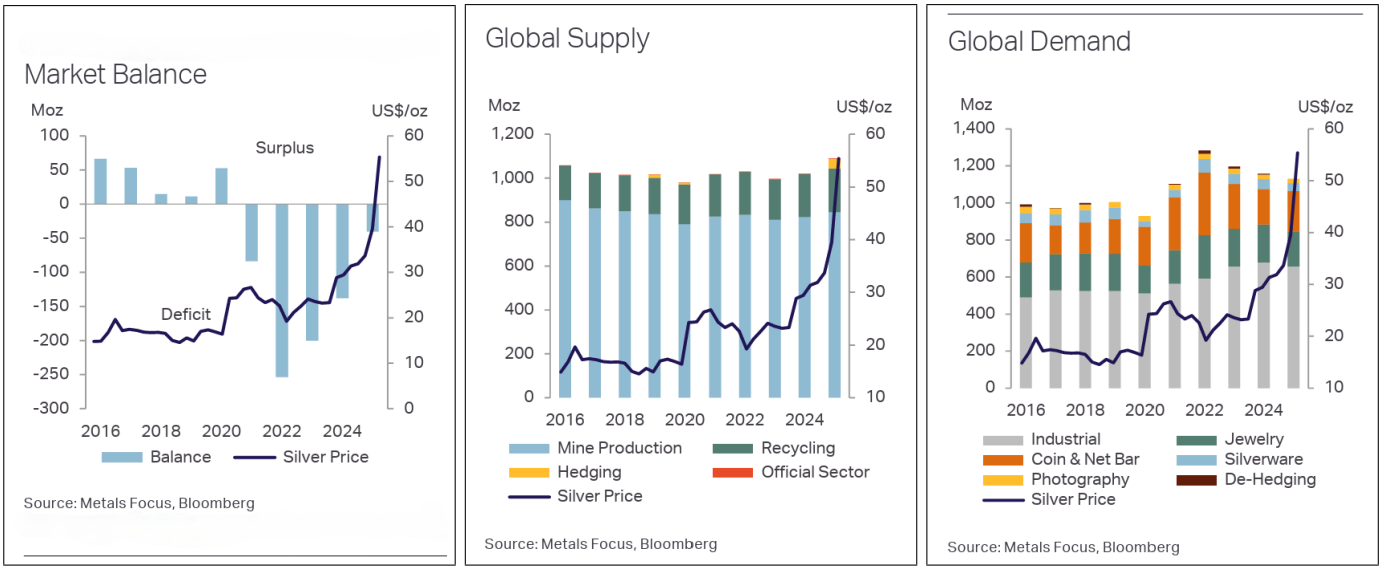
1. **Physical market stress and delivery risk**; Backwardation in futures and a sharp rise in silver lease rates, from a historical 0.3–0.5% range to ∼8%, indicate tightening physical availability and reducing confidence in paper-market settlement. For industrial users, access to metal now outweighs its cost.
2. **China's refined silver export controls**; China has placed roughly 60–70% of global refined silver exports under policy control, materially tightening supply available to global markets.
3. **Sovereign accumulation spillover**; While central banks are accumulating gold rather than silver, de-dollarisation efforts with BRICS initiative with "The Unit", and reserve diversification are pushing private and quasi-sovereign actors toward silver as a secondary monetary hedge. In a structurally deficit market, this incremental demand has an outsized impact on available supply.
4. **Energy-driven industrial demand**; To satisfy AI and clean-energy power needs, the world is rapidly restarting and expanding nuclear capacity. This creates a long-term, underestimated source of silver consumption.

Despite the fundamentally positive demand outlook, many analysts also caution high volatility. Some analysts even project silver spiking towards \$150 if central banks continue to buy silver as strategic reserve. Future prices will no longer be dictated solely by simple supply–demand balances, but by the interplay of geopolitics, monetary policy, and strategic reserve decisions. For silver-hungry industries, the greater risk is loss of access, not higher prices.

3 Quantifying the Deficit

Silver has now recorded five consecutive years of market deficits. 2024 deficit: ∼150 Moz, despite a 2% rise in supply and a 3% decline in demand. As seen in Figure 1, 2025 recorded deficit: 40.3 Moz, the fifth consecutive annual, where supply rose 7% YOY while demand eased marginally 2%. Although the annual deficit is expectedly narrowed in 2025 versus the 2022 peak, the market remains meaningfully undersupplied. Cumulative deficits since 2021 total 762 Moz — equivalent to nearly one full year of global mine production, an unparalleled situation. Above-ground inventories continue to be eroded rather than rebuilt. Importantly, price has not yet fully reflected physical tightness due to still-available inventories and macro-driven investor positioning. However, repeated annual deficits increase the probability of a more acute physical squeeze if inventories fall below functional levels.

* Independent analysis; views are personal.



(a) Balance (b) Supply (c) Demand

Figure 1: Silver Inventory: 5 years of sizeable deficits despite YOY supply increase and demand decrease.

4 EBITDA & Margin Impact

Solar PV is the most directly exposed segment. Silver has overtaken polysilicon as the largest single cost input in cell production, rising from 7 to 15% of COGS. From trough to peak in 2025 (silver: \$29 to \$84/oz, +190%), FOB China module prices rose only 15%, hence compressing margins. Manufacturers having absorbed the rest, lacked the ability to pass costs downstream due to structural overcapacity of the market. For a 10 GW manufacturer producing panels with 9 mg of silver per Watt, annual silver consumption is approximately 90 tonnes (2.9 Moz). Assuming market price \$0.125/W, estimated revenue is 10×10^9W $\times$ \$0.125/W = \$1.25B. At 8% margin, baseline EBITDA (E) is estimated at \$100M. Table 1 models three forward stress scenarios from the current spot price of \$75/oz, holding production volume and all non-silver costs constant.

R = revenue uplift capped at 10% of 1.25B = \$125M, ΔC = added silver cost. The column $E - \Delta C$ represents EBITDA under zero pass-through where no cost increase is recovered from buyers (plausible given persistent overcapacity in the PV market). The 10% pass-through column ($E - \Delta C + R$) applies the ceiling-capped revenue uplift R = \$125M, reflecting limited pricing power. Hedge cost is the annual premium for a 12-month call option at strike K, computed via the Black-Scholes with 12-month realised volatility.

Table 1: Forward Scenario Analysis: Silver Price Stress and Hedging Impact

Scenario	**Silver Price**	ΔP	ΔC	$E - \Delta C$	**Δ EBITDA**	**Δ Revenue**	$E - \Delta C + R$	**Δ EBITDA**	**Hedge Cost**
	K (\$/oz)	(vs \$75/oz)	$\Delta P \times 2.9$ Moz		**(0% PT)**	($\leq R$)		**(10% PT)**	**(call at K)**
Moderate	\$100	\$25	\$72.5 M	\$27.5 M	↓ 72.5%	\$125 M	\$152.5 M	↑ 52.5%	\$37.3 M
Severe	\$150	\$75	\$217.5 M	−\$117.5 M	EBITDA −ve	\$125 M	\$7.5 M	↓ 92.5%	\$15.8 M
Extreme	\$200	\$125	\$362.5 M	−\$262.5 M	EBITDA −ve	\$125 M	−\$137.5 M	EBITDA −ve	\$7.3 M

Assumptions: Spot \$75/oz, 9 mg/W silver intensity, \$0.125/W module price, 8% EBITDA margin. Hedge via call option, priced using Black-Scholes with $r = 4.5\%$ and 12-month realised historical volatility $\sigma = 65\%$.

The structural break-even, where $\Delta C = R$, i.e., cost increase equals the maximum recoverable revenue, occurs at \$118/oz. Beyond this, pass-through becomes insufficient irrespective of market conditions. A \$100 strike call, purchased today at \$37.3M, leaves the \$75 to \$100 exposure (\$72.5M) unprotected but caps all incremental cost beyond \$100, capping total exposure at approximately \$110M (ΔC + Hedge Cost)

against an unhedged exposure of $217.5M at $150/oz and $362.5M at $200/oz.

5 Strategic Recommendations

Despite a recent pull back in silver prices, the movement is macro-driven rather than a signal of deficit resolution. A bullish structural thesis remains intact, and the window to act before the next leg of the rally is narrow. For solar manufacturers, the following measures are critical to counter rising input costs.

1. **Financial Hedging**; It helps shift from a variable cost structure to partially fixed one. Buying a $100 call while selling a $60 put — nets zero upfront premium. Above $100, the call caps procurement cost. Below $60, the put obligates purchase at $60, still well below current spot. The manufacturer gives up participation in any sharp decline, but for an industrial buyer, cost predictability matters more.
2. **Supply Security**; Financial hedges protect the income statement but do not guarantee physical access In a market showing delivery stress, long-term offtake agreements with primary miners in Peru, Mexico, and Australia bypass the silver bottleneck created by Chinese export controls. Separately, 60–90 days of physical buffer inventory in segregated vaults should be treated as operational necessity. In a structurally tight market, leased metal may simply not be available when urgently needed, regardless of what you are willing to pay.
3. **Reducing Silver Intensity**; TOPCon currently consumes ~ 9 mg/W. Companies are investing in this R&D to target below 5 mg/W, potentially reaching 3 mg/W. They are effectively buying a structural cost hedge that no financial instrument can replicate.

References

[1] Metals Focus for The Silver Institute. *World Silver Survey 2026*. *link*.

[2] Bellini, E. (2026). Silver demand from PV industry expected to drop 19% this year. *pv magazine*. *link*.

[3] IndexBox. (2026). Silver Price Surge Squeezes Solar Module Margins in 2025–2026. *IndexBox Blog*. *link*.

[4] Saxo Bank. (2025). Silver's Breakout Year: From Monetary Hedge to Industrial Powerhouse. *Saxo Bank*, December 2025. *link*.

[5] CME Group. (2025). Battery Metals Hedging with Financially Settled Futures. *CME Group*. *link*.